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Parabolic avalanche scaling in the synchronization of cortical cell assemblies

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Abstract

Neurons in cortex synchronize their spiking in response to local and distant inputs. These synchronized assemblies are fundamental to cortex function, yet basic dynamical aspects about their size and duration are largely unknown. Using 2-photon imaging of neurons in superficial cortex of awake mice, we show that synchronized assemblies organize as scale-invariant avalanches that quadratically grow with duration. This quadratic expansion was found only for correlated neurons and required temporal coarse graining to compensate for spatial subsampling when network dynamics are critical, as demonstrated in simulations. The corresponding time course of an inverted parabola with exponent of $\chi = 2$ described avalanches of up to 5 s duration and maximized temporal complexity in the ongoing activity of prefrontal and somatosensory cortex and in visual responses of primary visual cortex. Our results identify a scale-invariant order in the synchronization of highly diverse cortical cell assemblies in the form of parabolic avalanches.

Significance Statement

The synchronization of cell assemblies is fundamental to many brain theories. Here we show such synchronization to grow according to an inverted parabola that maximizes temporal complexity. This quadratic scaling found for cell assemblies of highly diverse size and duration is in line with prediction for neuronal avalanches and the cortex being in a critical state.

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Introduction

Neuronal synchronization is fundamental to many theories on cortex function. Cortical neurons, by preferentially integrating recurrent activity from neighboring cells[1] and distant inputs, support various synchronization dynamics such as oscillations[2] and cascades, the latter in the form of waves[3,4], synfire chains[5,6], and neuronal avalanches[7,8]. Cortical cascades, in which a synchronized neuronal group facilitates synchronization in downstream neurons through select, converging connections, suggest a particular powerful mechanism to establish robust, yet flexible information processing in cortex[5-7]. However, both, the growth of cascades and their overall temporal profile, i.e., the number of neurons or spikes encountered over time, have been found in simulations to be variable, undermining information propagation in cortical networks[5,7,9-11]. This problem has been particularly prominent for neuronal avalanches[7], which represent highly diverse, scale-invariant cascades of neuronal activity predominantly found in the superficial layers of cortex. Avalanches are readily observed in the local field potential[12], selectively engage single neurons[13,14], and carry high information capacity[15,16]. Yet, it is currently not clear whether avalanches, when measured at the single cell level *in vivo*[8,17-19], do exhibit robust neuronal synchronization that unfolds in a predictive manner.

Traditionally, the hallmark of neuronal avalanches has been their scale-invariant spatiotemporal statistics quantified by power laws in size and duration[7] with exponents $\alpha \approx 3/2$ and $\beta \approx 2$, respectively. More recently, their temporal profile, i.e., how the size of an avalanche unfolds in time, has been suggested to discriminate between models of avalanche generation. Specifically, the scaling of their mean size with duration and their universal, duration-invariant temporal profile[20-22], can both be captured in a single scaling exponent χ. For models of avalanches lacking interactions, generated by noise, or those found near a 1st-order phase-

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transition, χ ranges between 1 – 1.5 and temporal profiles are non-parabolic, e.g. semi-circle and saw-tooth like[23,24]. In contrast, for avalanches that unfold according to a critical branching process, a close approximation for synchronized cascades[7,25], $\chi = 2$ and avalanche profiles are parabolic[11,20-22], similar to what can be found close to 2nd order phase transitions that fall into the directed percolation universality class[26] (see also [27]). This relationship has recently been shown for LFP-based avalanches in nonhuman primates[28], which suggests that avalanches describe rapid, scale-invariant unfolding of neuronal synchronization. However, when measured at the cellular level in the mammalian cortex, neuronal avalanches exhibited χ between ~1 – 1.3 with non-parabolic, even asymmetrical profiles[17,19,29,30], disenfranchising neuronal avalanches as a potential framework for cortical synchronization.

Contrary to those findings, we demonstrate here, using 2-photon imaging (2PI) in the cortex of awake transgenic mice, that $\chi = 2$ for neuronal avalanches which identifies a scale-invariant, temporal profile in the form of a symmetrical, inverted parabola that maximizes temporal complexity and increases temporal correlation in cortical population activity. These results establish a novel scaling relationship for cortical synchronization at cellular resolution in the form of parabolic avalanches.

Results

Synchronized assemblies in prefrontal cortex exhibit quadratic avalanche scaling in mean size vs. duration.

We studied neuronal synchronization in the prefrontal cortex of awake mice during resting and spontaneous locomotion using 2PI (Fig. 1). Chronic implantation of a prism[31] allowed us to simultaneously image from ~200 – 300 neurons within a window of 450 x 450 µm at a temporal

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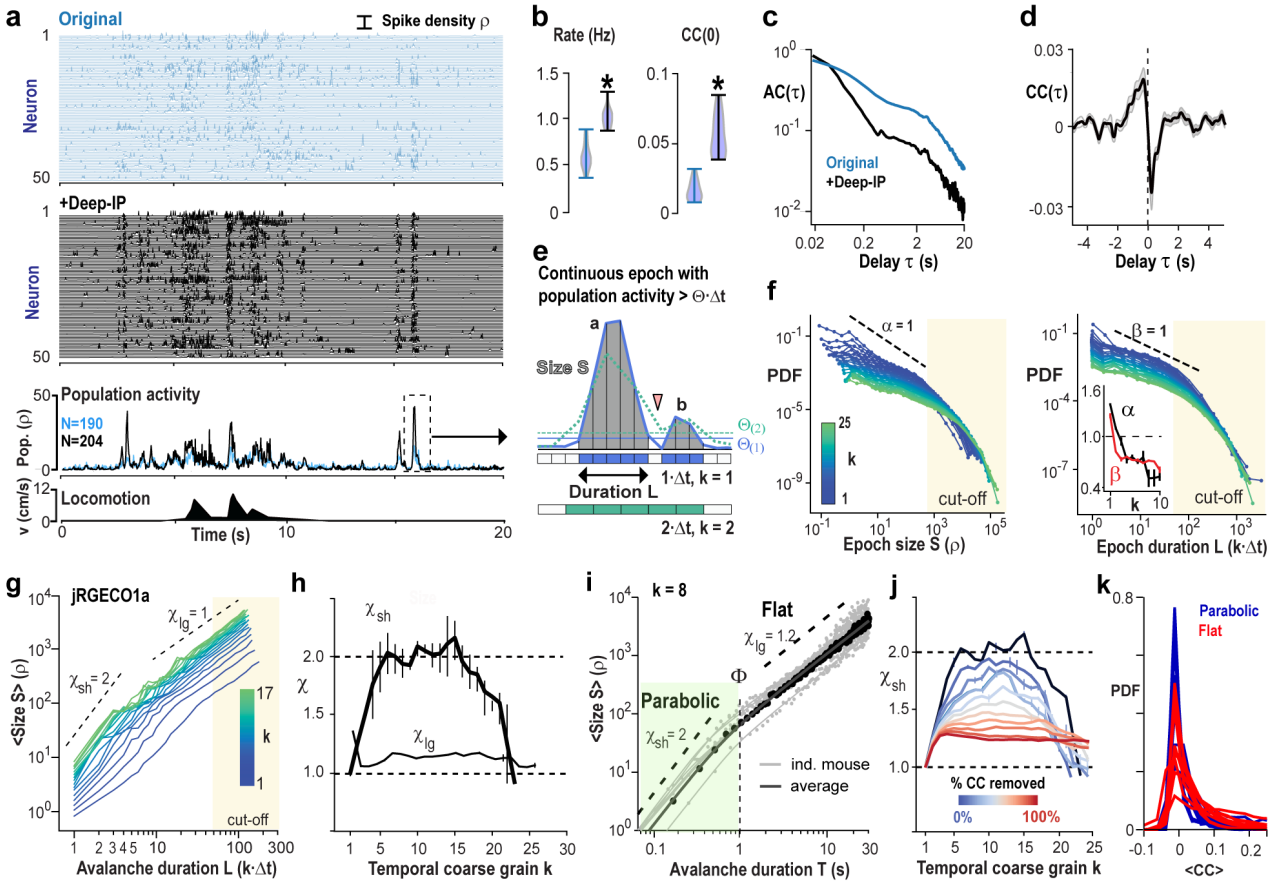


Figure 1| Neuronal synchronization exhibits avalanche scaling of $\chi = 2$ in prefrontal cortex. (**a**) Ongoing neuronal activity in mPFC/ACC of the awake mouse exhibits epochs of increased population activity (columnar organization; *original*) markedly more visible after deep-Interpolation (*+Deep-IP;*2PI, jrGECO1a). Spike density raster of 50 randomly selected neurons each (single mouse). *Middle*: Full population activity (*Pop*). *Bottom*: Instantaneous locomotion velocity. (**b**) Deep-IP increases neuronal firing (*Rate*) and pairwise correlation between neurons (*CC*). (**c**) Deep-IP sharpens population activity transients visible in a faster decay of the autocorrelation (*AC*). (**d**) Locomotion onset is preceded by population activity (*CC*). (**e**) Sketch of continuous epoch in population synchrony per *Δt* above threshold Θ(1) (*blue line*) and corresponding size *S* and duration *L* in multiples of time bins. The two suprathreshold epochs *a* and *b*, separated at *Δt* (*arrowhead*), will merge upon temporal coarse graining ($k = 2$), despite an increase in synchrony requirement per *Δt* (Θ(2); *broken green line*). (**f**) Suprathreshold epochs fulfill the criteria of neuronal avalanches. Epoch size *S* (*left*) and duration *L* (*right*) exhibit power laws robust to temporal coarse graining ($k = 1 – 25$; *color code*). Note cut-off regions starting at around $S > 100$ and $L > 50$ (*beige areas*). : Corresponding decreasing slopes $\alpha(k)$ and $\beta(k)$ cross the value of 1 with increase in *k*. (**g**) Rapid growth in mean size <S> vs. duration *L* for $k = 5 – 17$ for few-generation ($\chi_{sh} \cong 2$; $L = 1 – 4$), but not many-generation avalanches ($L = 10 – 20$; χ_{lg} remains $\cong 1.2$). (**h**) Summary of χ_{sh} and χ_{lg} as a function of *k*. (**i**) $\chi_{sh} = 2$ holds for 'parabolic' avalanches with durations $T < 1$ s at $k = 8$. In contrast, 'flat' avalanches have durations of $T > 2$ s. Values (*filled circles*) and corresponding fit (*solid line*) for each mouse and average. (**j**) $\chi_{sh} = 2$ requires high inter-neuronal correlations in the network and reduces to $\chi_{sh} << 2$ when neurons are made independent by random temporal shifts. (**k**) Pairwise correlations are similar during parabolic and flat avalanches. Single experiments for all mice shown. (*b – d, f – j*; averages for n = 5 mice; jRGECO1a). Values presented as mean ± SD.

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resolution Δt = 22 ms (~45.5 Hz frame rate) across the fissure in the contralateral, intact superficial layers of anterior cingulate and medial prefrontal cortex (ACC, mPFC; Fig. 1a; Suppl. Fig. S1). Machine-learning based deep-interpolation[32] (Deep-IP) markedly improved the extraction of spike densities from each neuron yielding significantly higher average firing rate and pairwise correlation among neurons (n = 5 mice; Fig. 1a, b; Suppl. Fig. S2). We summed activity across all neurons to obtain the time course of population activity. This time course exhibited rapid transients during which neurons showed coincident firing, i.e., spike synchronization in their activities, either spontaneously or prior to self-initiated locomotion (Fig. 1a, c, d).

We examined these epochs of synchronization systematically (1) by requiring the summed spike density within a time of Δt, i.e., population activity, to be larger than a threshold Θ, and (2) by concatenating successive suprathreshold population events (Fig. 1e). This commonly employed approach (e.g., [33]) to identify contiguous periods of significant network activity is sensitive to spatial subsampling and noise. Spatial subsampling prematurely terminates contiguous periods by missing neuronal activity, which we compensated for by systematically relaxing our concatenation criterion. Noise, on the other hand, reduces the ability to correctly identify suprathreshold periods. This error can be reduced by increasing Θ, which selects higher coincident neuronal activity in the network, i.e., higher synchronization. Accordingly, suprathreshold population events, identified at Δt, were concatenated at the temporally coarse grained resolution of $k \cdot \Delta t$, k = 1, …, k_{max}, $k_{max} < 40$. We further set $\Theta(k) = -2SD(k)$ of the z-scored distribution in number of epochs, which increases the absolute value of Θ with k (Fig. 1e; Suppl. Fig. S3a, b; see the Methods Section). We found that, independent of temporal coarse graining, epochs in population synchrony exhibited the hallmark of neuronal avalanches[7,8,12,14]

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with their sizes, *S,* i.e., summed suprathreshold spike densities, and durations, *L,* i.e., the number of time bins or generations per epoch, distributed according to power laws (Fig. 1f; Suppl. Figs. S3c; S4). The high diversity in size and duration of avalanches was quantified by the corresponding slopes α and β being smaller than 2 and rapidly crossing the value of 1 upon temporal coarse graining, demonstrating a broadening of the corresponding distributions despite robust size and duration cut-offs of $S > 10^3$ and $L > 50$, respectively (Fig. 1f; inset).

Next, we derived the scaling exponent, χ, which describes the power-law dependence between avalanche size and duration, visualized by plotting the mean avalanche size for avalanches of a given duration[20-22]. We found that χ for short, i.e., few-generation ($L < 5$) avalanches, $\chi_{sh,}$, significantly increased with temporal coarse graining, reaching maximal values around 2 (Fig. 1g, h). In contrast, χ for long, i.e., many-generation ($L > 10 – 30$) avalanches, χ_{lg}, remained close to 1 independent of the temporal scale at which epochs were observed (Fig. 1g, h). The transition Φ, obtained by fitting power law functions to these two regimes demonstrates that avalanches with absolute duration $T = L \cdot k \cdot \Delta t$ shorter than 1 s ($\Phi = 0.52 \pm 0.21$; n = 5) exhibited a quadratic, rapid growth in size as they unfolded in the network, which was not found for longer lasting avalanches ($T > \sim 2$ s; Fig. 1i). For simplicity, we define short-lasting avalanches with quadratic growth as ‘parabolic’ avalanches, in contrast to longer lasting ‘flat’ avalanches. Parabolic avalanches strongly depended on spatial correlations in the network. First, parabolic avalanches were abolished when neuronal time series were randomly shifted in time (Suppl. Fig. S2g) and secondly, χ_{sh} monotonically decreased to values lower than 2 with the percentage of spatial correlations removed (Fig. 1j) or percentage of uncorrelated spikes introduced (Suppl. Fig. S5a). Our approach was robust to subsampling given that up to a 50% reduction in our natural sampling fraction, *f*, still allowed us to robustly recover $\chi_{sh} \cong 2$ due to a

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concomitant increase in minimal requirement for coincident activity Θ (Suppl. Fig. S5b). Yet pairwise correlations between neurons were similar during parabolic and flat avalanches, suggesting parabolic avalanches exhibit distinct synchronization features of the neuronal network (Fig. 1k).

The finding of parabolic avalanches was robust to changes in Θ. As Θ increases, the number of avalanches increases rapidly because previously contiguous epochs become separated in time, whereas large sections of population activity are missed for very high Θ and the number of avalanches drops rapidly. The corresponding functions are well fitted by a lognormal distribution which shows large increases in avalanches when $\Theta(k)$ increased from -2SD(k) to -1SD(k)). The robustness to the choice of $\Theta(k)$ and levels of population activity is also in line with $\chi_{sh} \cong 2$ when randomly reducing our sampling fraction (cp. Suppl. Fig. S5b). Suppl. Fig. S3c demonstrates in density plots that $\chi_{sh} \cong 2$ over an order of magnitude of change in Θ.

The identification of $\chi_{sh} \cong 2$ was also robust to different forms of thresholding. Thresholding of an activity time series introduces a systematic error in the scaling estimate of χ when subthreshold values are included or excluded[34]. Our analytical derivation (Suppl. Info. S6) and corresponding data analysis demonstrates $\chi_{sh} \cong 2$ to be within the boundaries of both approaches (Suppl. Fig. S6a, b) and to be robust over a large range of thresholds (Suppl. Fig. S6c). Separating population activity into quiet rest and locomotion periods did not change our finding of $\chi_{sh} \cong 2$ despite a significant increase in neuronal firing rate when mice were spontaneously running (Suppl. Fig. S7).

We repeated our findings using GCaMP7s, which has a slower decay time constant and less tissue penetration capability for 2PI compared to jRGECO1a[35] (Fig. 2a – d; n = 3 mice; Δt = 22 ms). To focus on population synchrony without applying Deep-IP, we removed weakly

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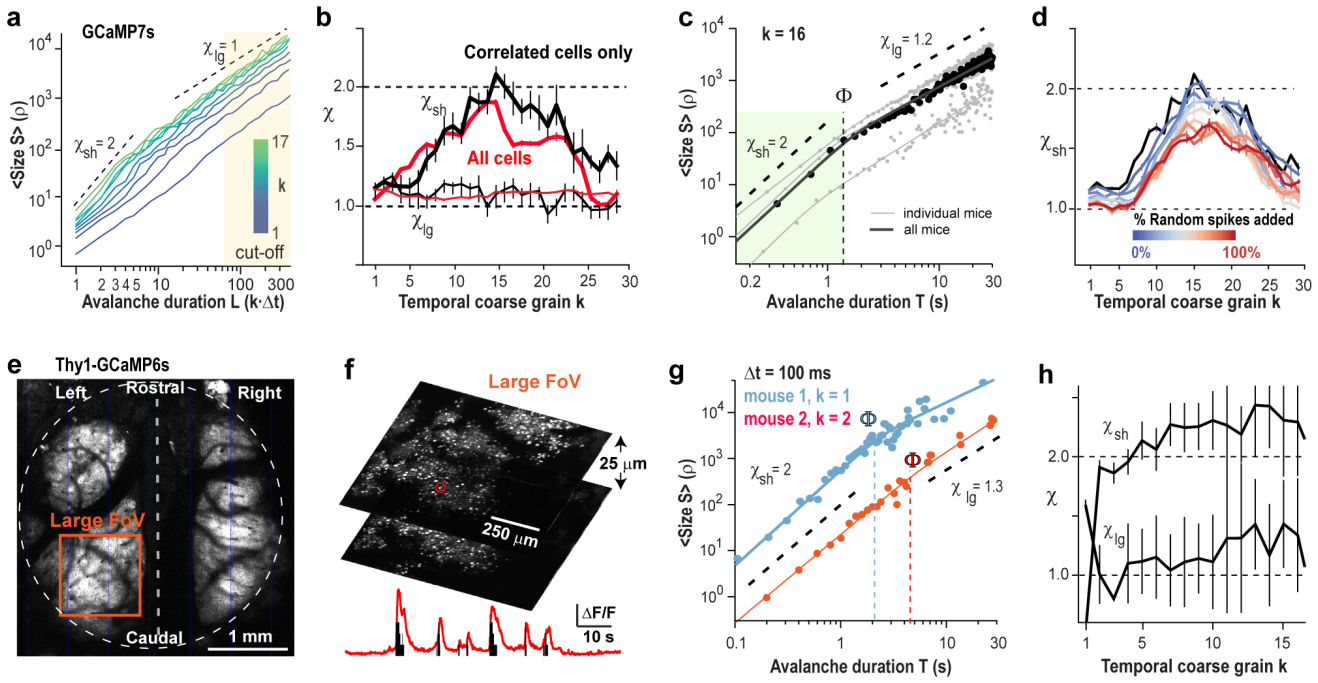


Figure 2| Avalanche scaling of $\chi = 2$ for ongoing neuronal synchronization is robust to numerous experimental conditions. (**a - d**) Ongoing activity in PFC/ACC using GCaMP7s and synchronized neurons (n = 3 mice). (**a**) $\chi_{sh} \cong 2$ is identifiable in the subpopulation of synchronized neurons for k =16. (**b**) Average value of χ_{sh} (*thick line*) and χ_{lg} (*thin line*) as a function of k for all (*red*) and correlated neurons only (*black*). (**c**) $\chi_{sh} \cong 2$ holds for avalanches of duration 0.3 – 1.8 s at k = 16. (**d**) Addition of random spikes, which mimics the presence of noisy, non-synchronized neuronal activity degrades the capacity to recover $\chi_{sh} \cong 2$ (GCaMP7s). *Black dashed lines*: Guide to the eye. Values presented as mean ± SD. (**e - h**) Ongoing activity in somatosensory cortex measured with transgenic mice intrinsically expressing GCaMP6s imaged using a mesoscope. (**e**) Overview of mesoscopic imaging of large brain area. Note expression of GECIs obscured by large blood vessels. *Broken line*: midline. *FoV*: Large (*orange*) Field of View. (**f**) High-resolution image of FoV in *e* and corresponding period of spontaneous fluctuation in relative fluorescence (*red*; au) and spikes (*black*) in a single neuron (*red circle*). (**g**) Scaling regime of χ_{sh} increases with FoV. For large FoV (*orange*), $\chi_{sh} \cong 2$ holds over for avalanches of duration 0.1 – 5 s at k = 1-2 (n=*2* mice; 10 min). (**h**) Corresponding average value of χ_{sh} and χ_{lg} as a function of k.

correlated neurons after z-scoring each neuron's correlation with the population activity (Suppl. Fig. S8a – d; 10 – 30% of neurons per recording; see the Methods section). In line with our findings using jrGECO1a, suprathreshold epochs in population activity exhibited the hallmark of avalanches (Suppl. Fig. S8e – g). Temporal coarse-graining revealed parabolic avalanches at k = 15 with duration of ~0.3 – 1.8 s that demonstrated $\chi_{sh} \cong 2$ and flat avalanches over longer

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durations with $\chi_{sh} \cong 1$. Again, $\chi_{sh} \cong 2$ was sensitive to contributions from uncorrelated cells or shuffling (Fig. 2d; Suppl. Fig. 8h), whereas flat avalanches were independent of temporal coarse-graining.

Our findings suggest that the two categories of parabolic and flat avalanches might arise from experimental shortcomings in accurately identifying neuronal avalanches. Sizes and durations of flat avalanches overlap with the cut-off regimes in the corresponding duration and size distributions (Figs. 1g, Fig. 2a; Suppl. Fig. 8e, f). Such cut-offs reflect finite-size effects from the field of view (FoV), which greatly impacts avalanche measures[36]. Indeed, when employing mesoscope imaging in somatosensory cortex of Thy1 transgenic mice expressing GCaMP6s (Fig. 2e, f), the maximal duration of parabolic avalanches correspondingly increased when increasing the FoV (Fig. 2g, h; Suppl. Fig. S9).

Our results establish that ongoing neuronal synchronization in frontal and somatosensory cortex organizes as parabolic avalanches with rapid, quadratic expansion in synchrony over time ($\chi_{sh} \cong 2$) for up to 5 s of duration over ~1 mm^2 of cortex.

Temporal coarse graining recovers $\chi = 2$ for critical branching process under subsampling conditions.

Next, we use simulations to explore the hypothesis that temporal coarse graining can recover parabolic avalanche synchronization in critical networks that are incompletely observed, i.e., spatially subsampled. We chose a network of $N = 10^6$ binary neurons (80% excitatory, E; 20% inhibitory, I) and all-to-all connectivity (Fig. 3a) with excitatory and inhibitory connectivity matrices $W_{EE} = W_{IE} = J$ and $W_{II} = W_{EI} = g \cdot J$ that were constant. Specifically, we set *g* to 3.5 and obtained an E/I-balance that supported avalanche dynamics in the fully sampled system with

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186 critical exponents $\alpha = 3/2$, $\beta = 2$, and $\chi = 2$ approximating a critical branching process triggered

187 by external Poisson inputs[37] (Suppl. Fig. S10). We found that temporal coarse graining recovered

188 $\chi_{sh} = 2$ for few-generation avalanches over a wide range of subsampling conditions denoted by

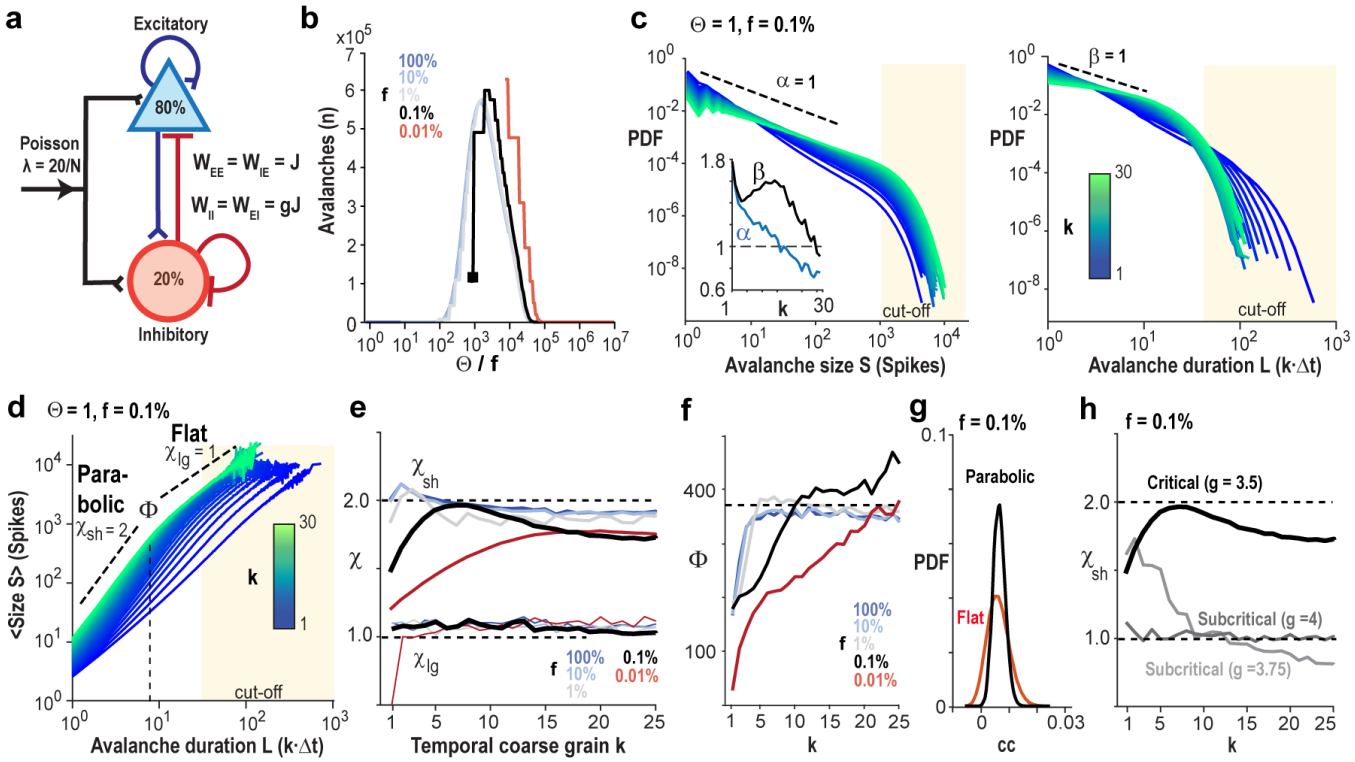


Figure 3| Temporal coarse graining recovers avalanche scaling of $\chi = 2$ under subsampling conditions when the network is critical. (**a**) Sketch of the neuronal network with $N = 10^6$ LIF neurons (80% excitatory, E; 20% inhibitory, I) and external Poisson drive of rate $\lambda = 20/N$ per time step. The E/I balance is controlled by the scalar g, which scales inhibitory weight matrices $W_{II} = W_{EI}$ as a function of excitatory weight matrices $W_{EE} = W_{IE} = J$. (**b**) Change in number of avalanches as a function of threshold Θ normalized by sampling fraction f ($T = 10^8$ time steps). (**c**) Power law in size (*left*) and duration (*right*) distributions for avalanches become shallower with temporal coarse graining k. Note cut-off regimes for $S >\sim 10^3$ and duration $L >\sim 50$. *Inset*: Corresponding slopes $\alpha(k)$ and $\beta(k)$. (**d**) Temporal coarse graining uncovers $\chi_{sh} = 2$ for short-duration avalanches ($L = 1 - 10$), whereas χ_{lg} remains ~1 – 1.2 for long-duration avalanches ($L > 10$). (**e**) Summary of change in χ_{sh} and χ_{lg} with k for 5 sampling fractions f. A decrease in f requires a higher k to recover $\chi_{sh} = 2$. Note failure of recovery for very low f. χ_{lg} does not depend on k. (**f**) Temporal coarse graining recovers avalanches up to the finite-size cut-off of $\Phi \cong 400$ time steps. Note plateau in maximal scaling range for $\chi_{sh} = 2$ plotted in simulation time steps as a function of k. (**g**) Neuronal cross-correlations are similar during parabolic and flat avalanches. (**h**) Temporal coarse graining recovers $\chi_{sh} = 2$ for critical network dynamics, but fails for subcritical dynamics. Note that weak subcritical ($g = 3.75$) and critical conditions ($g = 3.5$) can exhibit similar χ_{sh} at the original temporal resolution yet diverge with temporal coarse graining. *Broken, black lines*: visual guide to the eye. Averages over 10^8 simulation time steps.

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the sampling fraction, f = [0.1%, …, 100%] and population activity threshold, Θ = [0, …, <20,000 spikes per time unit]. We exemplify these findings in Fig. 3b – f using a sampling fraction of f = 0.1% and Θ = 0, i.e., the minimum requirement of 1 spike per time unit. In line with our experimental findings, temporal coarse graining of subsampled avalanches reduced the power law slopes in size and duration and recovered the mean size vs. duration power law relationship with $\chi_{sh} \cong 2$ for few-generation avalanches (Fig. 3e). In general, we found that sampling of fewer neurons (low f) and less sensitivity (high Θ) required more temporal coarse graining to recover $\chi_{sh} \cong 2$ (Fig. 3e). We note that successive avalanches are uncorrelated in the model, which reduces χ_{sh} for very large k independent of the total simulation time (Suppl. Fig. S11). As expected, $\chi_{lg} << 2$ for many-generation avalanches, which, similarly to our data, is located in the cut-off of the corresponding size and duration distribution (Fig. 3c, d).

The model also demonstrated that the scaling range Φ for which $\chi_{sh} \cong 2$ reflects a finite-size effect that is reliably recovered within the accuracy of the temporal coarse grain (Fig. 3f). As found for our 2PI data, noise reduces the recovered maximum value of $\chi_{sh} \leq 2$ without shift in k (Suppl. Fig. S12) and pairwise correlations are similar during parabolic and flat avalanches (Fig. 3g). Importantly, our simulations show that the recovery of $\chi_{sh} \cong 2$ under spatial subsampling is only possible when the network exhibits critical dynamics, whereas χ_{sh} remains $\cong 1$ for subcritical dynamics regardless of temporal coarse graining (Fig. 3h; Suppl. Fig. S13). Our simulations support the view that our 2PI data represent spatially subsampled activity of critical dynamics in cortex.

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209 **Evoked synchronization in primary visual cortex exhibits $\chi = 2$ avalanche scaling.**

210 Evoked visual and auditory responses in primary visual (V1) and auditory (A1) cortex have been

211 found to organize as neuronal avalanches exhibiting power laws in size and duration

212 distribution[17,18]. Yet, the relationship between the duration of evoked avalanches and their mean

213 size, χ , has been reported to be between 1 – 1.3 when measured at typical 2PI frame rates of,

214 e.g., 33 Hz[17,19]. We studied χ for neuronal responses in superficial layers of V1 to large field

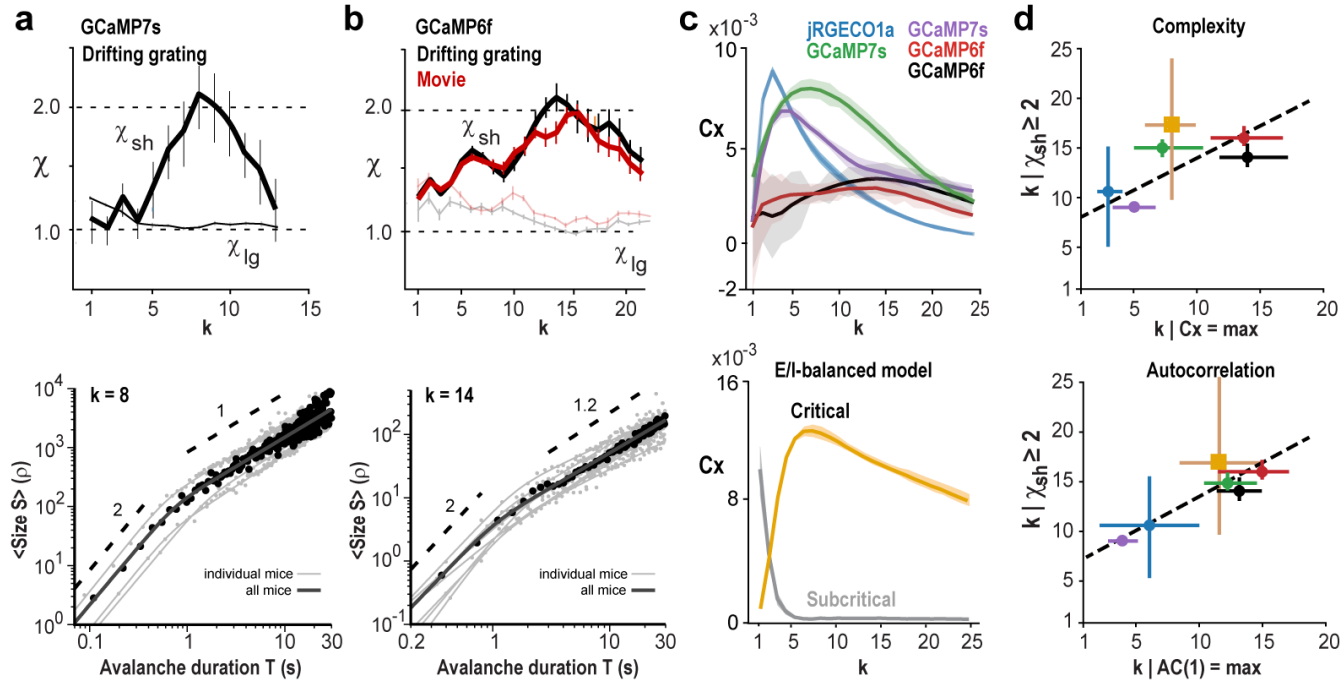


Figure 4| Visually evoked neuronal assemblies in V1 exhibit avalanche scaling of $\chi = 2$ at which complexity and temporal correlations are maximized. (**a**) *Top*: Temporal coarse graining uncovers $\chi_{sh} \cong 2$ for visually evoked, few-generation avalanches ($L = 1 – 4$). In contrast, many-generation avalanches ($L = 10 – 20$) exhibit $\chi_{lg} \sim 1$ at any temporal coarse graining (n = 3 mice; 45.5 Hz; GCaMP7s; +Deep-IP). *Bottom*: $\chi_{sh} = 2$ holds for evoked avalanches of duration T ~ <1 s for $k = 11 \pm 8$. (**b**) *Top*: Recovery of $\chi_{sh} = 2$ in V1 during viewing of drifting gratings (*black*) or movies (*red*; 33 Hz, GCaMP6f; n = 6 mice; Allen Institute data set). *Thin lines*: $\chi_{lg}(k)$. *Bottom*: $\chi_{sh} = 2$ holds for avalanche durations ~0.3 – 2 s for $k = 13 \pm 4$. (**c**) Peak in pattern complexity (*Cx*) with temporal coarse graining (Pattern depth $D = 5$, thresholded population activity). *Top*: Average per experimental condition. *Bottom*: In the E/I-model, pattern complexity peaks in subsampled, critical dynamics (g = 3.5; $P = 0.1\%$, $\Theta = 1$) with temporal coarse graining, but monotonically decreases for subcritical dynamics (g = 3.75; Pattern depth $D = 5$). (**d**) Temporal coarse graining at which $\chi_{sh} \geq 2$ maximizes pattern complexity (*top*) and temporal correlations (*bottom*). Averages over all experimental conditions and model. Error bars indicate range of k for which $\chi_{sh} \geq 2$ and 90 – 100% of maximal AC(1) or *Cx* respectively at baseline of $k = 1$ (R = 0.61 for *Cx*; R = 0.9 for AC(1)). For color code see *c*. *Thin solid lines*: Visual guides. *a*, *b*, *c*: mean ± SD.

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gratings drifting in 8 directions using GCaMP7s and in conjunction with Deep-IP and temporal coarse graining (Suppl. Fig. S14a – d; n = 3 mice). In line with our findings for ongoing activity, temporal coarse graining recovered $\chi_{sh} \cong 2$ for few-generation avalanches of durations T = ~0.1 – 1.6 s (Fig. 4a). These parabolic avalanches were abolished by trial-shuffling demonstrating their dependence on spatial correlations within each trial (Suppl. Fig. S14e – g). We then extended our analysis to the publicly available Allen-Institute data set on V1 evoked responses in superficial layers of cortex of the awake mouse. This data set used GCaMP6f, which preferentially reports action potential bursts[38] equivalent to applying a high, local Θ in our analysis. Responses to drifting gratings as well as to movies demonstrated $\chi_{sh} \cong 2$ under temporal coarse graining for few-generation avalanches, which again was abolished after trial and temporal shuffling, respectively (Fig. 4b; Suppl. Fig. S14h). Our analysis confirms and extends the demonstration of $\chi \cong 2$ from ongoing activity to evoked parabolic avalanches during sensory processing.

Avalanche scaling of $\chi = 2$ maximizes temporal complexity and identifies a scale-invariant inverted parabola in cortical synchronization.

Next, we explored how the initially increase and subsequent decrease in χ_{sh} with temporal coarse graining related to the temporal organization of neuronal population activity. Accordingly, we quantified the temporal complexity in population activity by dividing the thresholded population activity time course into $D \in 4, \ldots, 7$ successive temporal bins of duration $k \cdot \Delta t$. For each pattern depth, D, and coarse graining factor k, we calculated the pattern complexity, Cx, quantified as the number of different temporal sequences of length D (Ref.[39]; see Methods). We found that Cx peaked at k for which $\chi_{sh} = 2$, for both ongoing and evoked activity independent of pattern depth (Fig. 4c, d; Suppl. Fig. S15a, b). Our experimental findings were confirmed in the E/I-balanced

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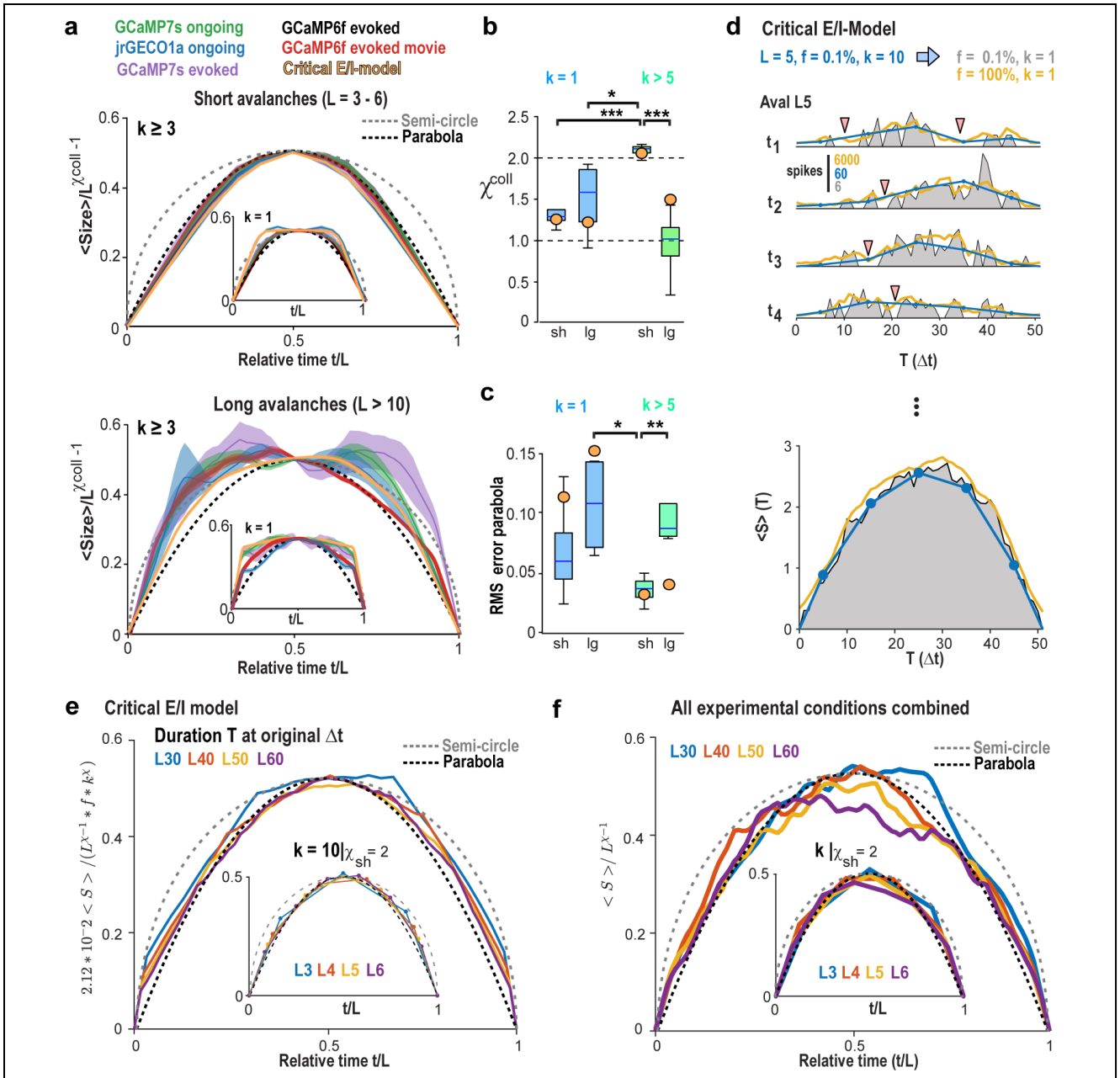


Figure 5| Spontaneous and evoked neuronal assemblies follow the universal temporal profile of an inverted avalanche parabola with scaling collapse of $\chi = 2$. (**a**) An inverted parabola describes the temporal profile of avalanches at $\chi_{sh} = 2$. *Top*: For short-lasting avalanches ($L = 3 – 6$; $k > 5$), an inverted parabola is recovered from the flattened, mean collapsed profiles present at $k = 1$ (*inset*). *Bottom*: Many-generation avalanches ($L = 10 – 20$) exhibit a flattened profile at $k = 1$ (*inset*) that differs from an inverted parabola even after temporal coarse graining. Average profile ± SD. Avalanches were pooled from all recordings from all mice corresponding to the respective experimental conditions (*color*). The mean activity within an avalanche normalized by the number of generations L to the power of χ^{coll}-1 is plotted for each relative time step, t/L. Note the match for E/I-balanced model with critical dynamics and subsampling ($P = 0.1\%$, Θ=1), which demonstrates recovery of critical exponent $\chi_{sh}^{coll} = 2$, and inverted-parabola collapse for short-lasting avalanches, but not otherwise. (**b**) Scaling exponent for profile collapse, χ^{coll}, is close to $\chi_{sh}^{coll} = 2$ for coarse-grained, short-lasting avalanches and significantly lower $\chi_{lg}^{coll} < 2$ for many-generation avalanches.

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(Continued from previous page Fig. 5) Summary statistics for all 5 experimental conditions from *a*. *Orange circles*: Values from the critical E/I-model with SD < symbol size. (**c**) Short-lasting, coarse grained avalanches exhibit profile closest to a parabola. (**d**) In the critical model, population activity sequences t_i defined by $L = 5$ at $k = 10$ (*blue*) exhibit interruption of contiguous suprathreshold epochs due to subsampling (*grey*), not found in the fully sampled model (*yellow*). *Bottom*: The inverted parabolic shapes of $L = 5$, $k = 10$ epochs (*blue*) match corresponding time averages from subsampled and fully sampled condition. (**e**) Summary for L30, L40, L50, and L60 at original Δt and corresponding profile for avalanches at $k = 10$ for which $\chi_{sh}^{coll} = 2$. (**f**) Corresponding analysis pooled over all experiments. *b*, *c* One-way ANOVA. d, e, f averages only.

complexity with temporal coarse graining. Our simulations also demonstrated that lower sampling fractions require higher temporal coarse graining to recover peak complexity (Suppl. Fig. S15c), which is in line with our findings for χ_{sh} in the data (*cf*. Fig. 3e). This increase in temporal complexity was in line with our finding that the delayed auto-correlation AC(1) peaked for parabolic avalanches, but not all avalanches or flat avalanches, at the k for which $\chi_{sh} = 2$ (Fig. 4d; Suppl. Fig. S16).

Theory and experiment predict that the slope value of $\chi_{sh} \cong 2$, consistently found in our experiments, predicts scale-invariant, inverted parabolic profiles for avalanches, which can be collapsed with an exponent $\chi^{coll} \approx 2$ (Ref. [20-22]; see Methods). We show in Figure 5 the corresponding collapsed avalanche profiles for few- and many-generation avalanches for all experimental conditions and the critical model. Indeed, an inverted parabolic shape was only found for temporal coarse graining k that maximized χ_{sh} with corresponding values of χ^{coll} close to 2 (R between χ_{sh} and χ^{coll} of 0.97 ± 0.03; Fig. 5a – c). In contrast, short-generation avalanches at temporal coarse graining that did not exhibit $\chi_{sh} \cong 2$ as well as many-generation avalanches revealed a flattened profile that deviated from a parabola. We note that parabolic avalanches (L = 3 – 6; $\chi_{sh} \geq 2$) did not recur regularly, which separates them from oscillatory activity (Suppl. Fig. S17). Using the temporally coarse-grained time segments of parabolic avalanches, we recovered their corresponding inverted parabolic shape in the average at the original temporal resolution of

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257 Δt for the model and our data (Fig. 5d – f; Suppl. Fig. S18a, b). In the model, this approach can
258 be used to recover segments of parabolic and flat avalanches in the subsampled system at
259 original temporal resolution (Fig. S18c).
260

261 ## Discussion

262 We identified an inverted parabolic profile in the synchronization of neuronal groups of
263 prefrontal and primary sensory cortex. This symmetrical profile exhibits scale-invariance over
264 hundreds of milliseconds and seconds and maximizes the temporal complexity of neuronal
265 synchronization during quiet resting and spontaneous locomotion as well as for visually evoked
266 responses under a variety of different recording conditions.

267 The synchronized groups reconstructed here at cellular level for superficial layer 2/3
268 reside robustly within the framework of neuronal avalanches, defined by power laws in neuronal
269 group size and duration. These power laws, when interpreted within the framework of self-
270 organized criticality (SOC;[40]), suggest a critical state of cortex that supports diverse and
271 potentially system-wide propagation of synchronized activity[13,16,41]. This interpretation has been
272 challenged recently by alternative models of avalanche generation utilizing balanced noise in the
273 absence of neuronal interactions[23,42] and noise-induced fractures near a discontinuous, 1st order
274 phase transition including hysteresis[43], recently summarized as self-organized bi-stability
275 (SOB;[24,44]). Our finding of a scaling exponent of $\chi = 2$ that is sensitive to spatial correlations
276 does not support these models, for which χ ranges between 1 – 1.5 and for which shapes are non-
277 parabolic, closer to semi-circles or saw-tooth like (e.g.[23,24,34]). Our results also clarify recent
278 experimental findings on neuronal avalanches in cortex with a scaling exponent χ close to ~1.3
279 (Ref.[17,19,29]). By introducing temporal coarse-graining in combination with deep-interpolation

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based de-noising, we overcame these experimental limitations and demonstrated that indeed neuronal avalanches in cortex reflect synchronized neuronal groups with a scaling exponent of $\chi = 2$.

Our findings are in line with avalanche generating mechanisms that reside close to a 2nd order phase transition, belonging to the directed percolation universality class, e.g., a critical branching process (see also Fig. 3)[7,10,45-48] and specific types of processes found in intermittent magnetization near a critical state[21]. Our results are also in line with experimental recordings on 'Barkhausen noise' by Sethna et al.[21] who demonstrated power laws in size and duration with $\alpha \approx 1.5$ and $\beta \approx 2$, respectively, and an inverted parabolic profile with $\chi = 2$ for short avalanches, whereas a flattened profile and $\chi = 1$ was found for long avalanches[20] in line with recent findings in simulations of critical networks[27].

Our simulations also show that $\chi = 2$ is only possible at criticality, regardless of coarse graining level or threshold applied. In critical models, under conditions of complete 'separation of time scales', subsampling can be compensated for by simply temporally integrating up to the finite-size cut-off of the system[49], a solution which is not applicable for experimental data. Extending beyond cortex, recent whole-brain zebra fish analysis demonstrated 'crackling noise'[21] to describe 3-dimensional propagation of activity with exponents $\alpha \approx 3$, $\beta \approx 2$, much steeper than in the current analysis, and estimates of χ to range between 1.6 – 1.8 (Ref. [50]).

It can be shown that the critical power law exponents for avalanche duration, size and duration-size scaling are related by the formula $(\beta-1)/(\alpha-1) = \chi$ at the critical point, which was found for certain models[21,22]. We consistently found that temporal coarse-graining preserves the power laws in avalanche size and duration distributions under subsampling conditions, however, it does so with increasingly shallower slopes[14] introducing a singularity in $(\beta-1)/(\alpha-1)$ when $\alpha = 1$

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(cf. Fig. 1f; Fig. 3c; Suppl. Fig. S8g). This singularity was even present in our subsampled model, despite the fully sampled model exhibiting critical exponents $\alpha = 3/2$, $\beta = 2$ which fulfilled the analytical prediction of $\chi = (\beta-1)/(\alpha-1) = 2$. We conclude that spatial subsampling, as shown here and reported previously[51] as well as finite-size effects from spatial windowing[52] prevent straight-forward ratio calculations of the critical exponent χ from slope ratios obtained for size and duration distributions. We also note that this ratio relationship was derived in the absence of any external drive and with infinite separation of time scales, which differs from brain activity.

The coincident firing of neurons within a temporal window Δt was used to identify brief periods of synchronous population activity. Our minimal threshold increased with temporal coarse graining, which equates to requiring an increasingly higher minimal number of coincident spikes within the population of neurons. This synchronization requirement applies to each generation or $k \cdot \Delta t$ and thus, the overall duration of an avalanche does not affect the definition of minimal synchronized activity. Secondly, our randomization controls demonstrated that χ_{sh} rapidly drops below 2 and cannot be recovered by temporal coarse graining. In these controls, we reduced the percentage of coincident spikes by either adding random spikes or removing correlated pairs of neurons for which correlation was measured at zero-time lag, thus quantifying coincident firing. Accordingly, we conclude that our results capture synchronized neuronal population activity in the cortex.

The scale-invariant, inverted parabola represents a novel form of synchronization that complements commonly considered synchronization dynamics such as oscillations, waves, and synfire chains. The temporal profile of oscillations, while similar to an inverted parabola, would exhibit dominant durations and repeat regularly. Both aspects are not supported by our findings,

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which suggest scale-invariance without prominent periodicity. Propagating wave fronts that traverse our field of view would result in more flattened profiles and $\chi \approx 1$ like what we found for many-generation avalanches. A similar argument holds for synfire chains with their typically assumed constant layer width, which approximates extent-limited propagating wave fronts. Waves or synfire chains limited to the FoV would not be expected to unfold according to a scale-invariant, inverted-parabola.

Our finding of a symmetrical profile across numerous experimental conditions and spatial resolution such as cellular spiking and the mesoscale based LFP suggests profile symmetry to be an important constraint for theories on brain synchronization. Such robust symmetry is unexpected. Avalanche propagation in the brain exhibits robust functional connectivity that is heavy-tailed or small-world[53,54], which simulations and theory suggest to support asymmetric profiles of critical network cascades[11]. Asymmetric avalanche profiles have also been reported for Barkhausen noise[55] when experimentally applying external forces, which, again would predict asymmetric profiles for, e.g., sensory-driven, neuronal avalanches.

The reconstruction of local synchronization reported here from cellular 2PI data approximates local synchronized activity captured in the local field potential[12,14]. Accordingly, our identification of $\chi = 2$ for 450 x 450 µm FoV of superficial cortex using 2PI is in line with our recent demonstration of $\chi = 2$ for LFP-avalanches in superficial layers of nonhuman primates over an area that is 100 times larger[28].

We propose that the scale-invariant, inverted parabola in synchronization complements alternative frameworks of synchronization and potentially circumvents challenges presented by other measures. It is now well established that neuronal avalanches and oscillations co-exist *in vitro* and *in vivo* [28,56,57], a co-emergence dependent on e.g., the E/I-balance[33,58-60]. However,

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oscillations emphasize phase-locked firing among neurons, which limits the number of patterns that can be phase-coded per cycle[61,62]. Similarly, while the spatial unfolding of neuronal avalanches favors nearby spatial sites in the aggregate[36], travelling waves [3,63], given their spatial compactness, might limit spatial selectivity and simultaneous occurrence of waves within a brain region[64]. The relationship between avalanches and synfire chains is less clear. Synfire chains, by recruiting specific groups of neurons in each feed-forward layer[5,6,9], in principle could be highly adaptive and selective, however, are difficult to stably embed in recurrent networks. In contrast, neuronal avalanches represent selective neuronal participation in propagated synchrony that maximizes the information that can be stored in highly diverse avalanche patterns[15,16,65,66]. Our results and simulations demonstrate a temporal 'gestalt' of highly variable cell assembly synchronization in line with predictions of critical dynamics in cortex.

Methods

Overview on data sets analyzed

To demonstrate the robustness of our scaling results, we used 6 different datasets surmising 4 different probes to monitor ongoing activity in superficial layers of frontal cortex (ACC/mPFC) and evoked activity in superficial layers of primary visual cortex.

Ongoing activity in contralateral ACC/mPFC monitored with GCaMP7s and jRGECO1a

Mice (C57BL/6; Jackson Laboratories) were injected with a viral construct to express either GCaMP7s or jRGECO1a in cortical neurons using the Syn promotor. Chronic 2PI started after >2 weeks in the contralateral ACC/mPFC at an estimated depth of ~150 – 300 μm using a microprism. Recordings were collected over the course of several days from n = 3 mice with GCaMP7s expression (n = 19 recordings; 30 min each) and n = 5 mice with jrGECO1a

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expression (n = 25 recordings; 30 min each). Recordings were analyzed separately for each mouse.

Ongoing activity in somatosensory cortex using transgenic Thy1-mice expressing GCaMP6s

C57BL/6J-Tg(Thy1-GCaMP6s)GP4.3Dkim/J mice were obtained from Jackson labs (https://www.jax.org/strain/024275) and bred inhouse with C57BL/6J mice (Jackson Laboratory). Under a reversed 12:12 hr light/dark cycle with ad lib access to a running wheel, mice were group-housed until the day of surgery and single-housed thereafter. Recordings of 10 min of ongoing activity from n = 2 mice were performed over an area of ~1 mm^2 of cortical superficial layers.

Visually evoked activity in V1 monitored with GCaMP7s

Mice were injected with a mixture of a viral construct to express GCaMP7s in pyramidal neurons using the CaMKII promotor. Chronic 2PI started after >2 weeks in identified V1 at a depth of ~150 µm in response to drifting gratings from n = 5 mice over the course of several days.

Allen Brain Observatory Visual Coding dataset monitored with GCaMP6f

Two additional data sets on visually evoked V1 responses were analyzed from the publicly available Allen Brain Observatory Visual Coding dataset (https://observatory.brain-map.org/visualcoding). These data were collected using GCaMP6f in n = 8 transgenic mice (Cux2-CreERT2-GCaMP6f AI94) passively viewing drifting gratings and natural movies, imaged at 275 µm depth in V1 (~25 min recordings). For analysis, we used the deconvolved time series, extracted from 2-photon ΔF/F signal using an L0 regularization algorithm, made available through 'allensdk' by the Allen Institute. We limited our analysis to recordings from superficial layers in V1 that contained at least several hundreds of neurons and total stimulation time of many minutes. Data corresponding to 'drifting gratings' and 'natural movie 3' stimulation

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conditions were picked out of a longer recording using the 'allensdk' 'stimulus_table' object. For stimulation conditions with quickly varying stimulus, the 'stimulus_table' object was used to extract the start/end time for the entire stimulation epoch.

Animal surgery

All procedures were approved by the NIH Animal Care and Use Committee (ACUC) and experiments followed the NIH *Guide for the Care and Use of Laboratory Animals*. Mice were obtained from Jackson labs, bred inhouse with C57BL/6 backgrounds (Jackson Laboratory) under a reversed 12:12 hr light/dark cycle. Chronic 2PI imaging was performed using a head bar in combination with a cranial window implanted in adult (>6 weeks) mice. Implants consisted either of (1) a microprism/coverslip assembly to image through the medial wall of the contralateral ACC/mPFC following the procedure detailed in[31] or (2) a stack of circular glass cover slips using established protocols[67]. For ACC/mPFC recordings, 2 – 3 injections of virus (100 – 400 nL; <1 µL in total; 10^{13} vg/mL; pAAV/.Syn.NES-jRGECO1a.WPRESV40--AAV9, pGP-AAV-syn-jGCaMP7s-WPRE AAV9, Addgene) were administered into the hemisphere contralateral to the prism implant (+0.75 – 1.25 mm AP, 0.1 – 0.3 mm lateral, ~250 µm below the pial surface). To monitor V1 activity, the cranial window was placed centered at ~2.5 mm from the midline (right hemisphere) and ~1 mm rostral to the lambdoid suture.

For mesoscope experiments using transgenic Thy1-GCaMP6s mice, adult mice (>6 weeks) underwent a head bar surgery in combination with a slightly modified cranial window implant to allow for a larger area of imaging. In short, the window implant consisted of 3 layers of No.0 coverslips. The top coverslip (5 mm diameter) was used to close the craniotomy, whereas two smaller diameter cover slips (4 mm diameter) were used to gently fill the cavity between dura

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418 and the removed skull in order to prevent bone regrowth. This procedure resulted in clear
419 craniotomies over extended periods of time with an area of approximately 4 mm × 4 mm
420 accessible for imaging. The cranial window was centered above the midline at ~0 Bregma[68].
421
422 **Identification of V1 maps**
423 Retinotopic maps of V1 and higher visual areas (HVAs) were generated for all mice prior to
424 recording using published protocols[69,70]. Briefly, awake, head-fixed mice faced with their left eye
425 a 19" LCD monitor placed at 10 cm distance and tilted 30º towards the mouse's midline. Using
426 Psychophysics toolbox[71], contrast-reversing, spherically corrected checkerboard bars were
427 drifted across the screen vertically (altitude) and horizontally (azimuth) for each of the four
428 directions (30 repeats per direction). Simultaneous wide-field imaging (Quantalux, Thorlabs)
429 captured GCaMP7s fluorescence, which was averaged for each direction. Altitude and azimuth
430 phase maps were calculated by phase-wrapping the first harmonics of the 1D Fourier transform
431 for each of the four averages and subsequently subtracting the maps of the opposite directions[70].
432 Sign maps were generated by taking the sine of the angle between the gradients in the altitude
433 and azimuth maps and processed as previously described[69]. Borders were drawn around visual
434 area patches and overlaid onto anatomical reference images to identify V1.
435
436 **Visual stimulation and response measures**
437 Visual stimuli were prepared in MATLAB using the Psychophysics Toolbox[71] and delivered via
438 a monitor (Dell, 60 Hz refresh rate) placed ~25 cm in front of the contra-lateral eye of the mouse.
439 The stimulus was composed of moving gratings at 8 different directions presented for 1 s at
440 maximum contrast, 0.04 cycles per degree and 2 cycles per s. Stimuli were interspaced by gray

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screen (average luminance matched to stimuli) for 7 s. Each direction was presented 20 times in randomized order, for a total of 160 iterations. To confirm retinotopic identification of V1, we also calculated the direction selectivity index, DSI, for each neuron using the common formula $DSI = (R_p - R_o)/(R_p + R_o)$, where R_p and R_o are the responses to the preferred and opposite direction, respectively. Significance of DSI for each cell was assessed by comparing the values obtained from the original data with those obtained from shuffling the inter-spike intervals (data not shown).

2PI imaging, pre-processing pipeline and meta data collection

For standard 2PI, images were acquired by a scanning microscope (Bergamo II series, B248, Thorlabs) coupled to a pulsed femtosecond Ti:Sapphire 2-photon laser with dispersion compensation (Discovery, Coherent Inc.). The microscope was controlled by ThorImageLS software. The wavelength was tuned to either 940 nm or 1120 nm in order to excite GCaMP7s or jRGECO1a, respectively. Signals were collected through a 16× 0.8 NA microscope objective (Nikon). Emitted photons were collected through 525/50 nm (GCaMP7s) or 607/70 nm (jRGECO1a) band filters using GaAsP photomultiplier tubes. The field of view was ~450 x 450 µm. Imaging frames of 512×512 pixels were acquired at 45.527 Hz by bidirectional scanning of a 12 kHz Galvo-resonant scanner. Beam turnarounds at the edges of the image were blanked with a Pockels cell. The average power for imaging was <70 mW, measured at the sample.

For mesoscope imaging, images were acquired by a 2Photon Random Access Mesoscope (2P-RAM, Thorlabs) coupled to a pulsed ultrafast tunable femtosecond laser (Chameleon Discovery NX, Coherent Inc.). The Mesoscope was controlled by ScanImage software (ScanImage, Vidrio Technologies). The wavelength was tuned to 920 nm in order to excite GCaMP6s. The specimen

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was excited at NA = 0.6 and 2-photon signals were collected at NA = 1 using 4 separate GaAsP photomultiplier tubes for 2 channels and 2 planes of imaging. The field of view was ~1 x 1 mm^2. Imaging frames of 1024×1024 pixels (yields to 0.97µm per pixel in lateral direction) were acquired at ~10.5 Hz by a 12 kHz resonant scanner combined with a virtually conjugated Galvo scanner set. The total power delivered to the specimen for imaging was <70 mW, which was split approximately equally between the 2 planes. The dual-plane imaging adds a shadow of each plane onto the other. Redundant cells were removed by detecting cells located at the same coordinates (with a 2 µm tolerance) across the two planes that are highly correlated (> 0.7) and then selecting only the one with the highest signal.

The obtained tif-movies in uint16 format were rigid motion-corrected via the python-based software package '*suite2p*'[72]. Registered images were further denoised using machine-learning based, deep interpolation[32] (see below) and then semi-automatically processed by suite2p for ROI selection and fluorescence signal extraction. For each labeled neuron, raw soma and neuropil fluorescence signals (red for jRGECO1a; green for GCaMP6f/s, GCaMP7s) over time were extracted for each ROI. Spiking probabilities were obtained from neuropil corrected fluorescence traces ($F_{corrected} = F_{ROI} - 0.7*F_{neuropil}$) via MLspike by utilizing its autocalibration feature to obtain unitary spike event amplitude, decay time, and channel noise for individual ROIs.

Deep-Interpolation

Deep-Interpolation (Deep-IP;[32]) removes independent noise by using local spatiotemporal data across a noisy image stack of $N_{pre} + N_{post}$ frames to predict, or interpolate, pixel intensity values

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throughout a single withheld central frame. The deep neural network is a nonlinear interpolation model based on a UNet inspired encoder-decoder architecture with 2D convolutional layers where training and validation are performed on noisy images without the need for ground truth data.

After rigid motion correction, individual denoised frames were obtained by streaming one 60-frame (N_{pre} = N_{post} = 30 frames) registered, image stack through the provided Ai-93 pretrained model[32] for each frame to be interpolated. Results were similar compared to Ai93 models which underwent transfer-training using our GCaMP7s and jrGECO1a data sets (data not shown). At an imaging rate of ~45 Hz, these 60 frames correspond to a combined ~1.3 seconds of data surrounding the frame to be interpolated. To study the effect of Deep-IP, we omitted this step in our analysis of ongoing activity from our jRGECO1a recordings. In addition, we compared the additional effect of removing weakly population correlated ROIs (see Suppl. Fig. S2).

Locomotion speed

During imaging sessions, mice were head-fixed on a wheel on which they were free to run during collection of ongoing activity. The wheel was arrested during visual stimulation to reduce trial-by-trial variability. Locomotion speed was recorded via a custom-made photo diode sensor attached to the bottom of the recording platform. As the mice ran, the wheel turning made its teeth periodically block the IR light emitted by the sensor. This produced a square-wave pattern of reflected IR light on the sensor. Using wavelet-analysis and thresholding, an instantaneous locomotion speed was reconstructed at frame rate resolution. Population activity was correlated with locomotion for ongoing activity using instantaneous or 1-s smooth locomotion speed

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estimates. Cross-correlation functions were detrended, combined over experiments for each mouse, and averaged across mice.

Postprocessing pipeline

Uncorrelated cell removal

In order to focus on synchronized population activity, we identified those cells in the population that were uncorrelated with the overall neuronal population. This procedure proved necessary in data sets that were not denoised by deep-interpolation. In contrast, once our imaging data were denoised, the percentage of uncorrelated cells dropped to negligible values (see Suppl. Fig. S2).

For each cell i in an event raster, the cell's population correlation c_i was computed by taking the Pearson's R between the cell's activity, $r_i(t)$, and the all-except-i population activity timeseries $p(t) = \sum_{j \neq i} r_j(t)$. Then, a null population correlation $c'_{i,\tau}$ was computed as the Pearson's R between the all-except-i population vector and $r'_{i,\tau}(t)$, where $r'_{i,\tau}(t)$ is the cell i's activity time series shifted forward in time by τ frames, where the last τ frames are shifted to the start of the vector (circular shift). Null correlations were computed for τ in range $\tau \in [-100,100]$. From this a distribution of null correlations C' was formed, and from that a z-scored correlation value was calculated as $z_i = (c_i - \overline{C'})/\sigma^2_{C'}$. All cells with $z_i < 0.01$ were removed from the recording.

Continuous epochs of suprathreshold population activity

Continuous periods of population activity were identified by applying a threshold Θ on the population vector $p(t)$, the sum of the spike densities from all neuronal ROIs at a given time t, such that:

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$$p_\Theta(t) = \begin{cases} p(t) & p(t) > \Theta \\ 0 & p(t) \leq \Theta \end{cases}$$

For a given recording $p(t)$ and coarse graining value k, the dependence of the number N of epochs on the threshold Θ, N(Θ) was obtained for a range of thresholds $\Theta \in [\Theta_1, \Theta_2]$ such that Θ_1 was low enough that it removed no activity from the population vector and Θ_2 was high enough that it would remove all activity from the population vector. The function N(Θ) was typically well-approximated by a log-normal distribution and a corresponding fit yielded shape parameters μ and σ. The threshold used in the analysis for all recordings and coarse graining factor was chosen such that $\Theta = \mu - 2\sigma$ and estimated for each k.

We note that thresholding the population activity obtained from a set of spiking neurons was first applied by Poil et al.[33] in neuronal simulations to study neuronal avalanches. This method is similar to thresholding the local LFP (see e.g.[14]), the traditional approach in population-based analysis of neuronal avalanches, where it was shown that the amplitude of the local negative deflection in the LFP monotonically increases with the number of neurons firing within the vicinity of the microelectrode (e.g.[12]). Importantly, we chose Θ to be low so as to minimize potential errors in the estimate of χ[34] and thereby reduce the number of epochs obtained for our scaling analysis.

Temporal coarse graining

A temporal coarse-graining operation was applied to the thresholded population activity $p_\Theta(t)$. For a given temporal coarse graining factor k an ensemble of K different coarse-grained time series $p_k^0(\tau),\ p_k^1(\tau), \dots, p_k^{K-1}(\tau)$ was arrived at through the following method:

$$p_k^j(\tau) = \sum_{i=k\tau+j}^{k(\tau+1)+j-1} p_\Theta(i) \quad for\ \tau \in \{0, 1, \dots, \lfloor (T-j)/k \rfloor\}$$

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For each time series $p_k^j(\tau)$, epochs were extracted by finding pairs (τ_1, τ_2) such that $p_k^j(\tau_1) = 0$, $p_k^j(\tau_2) = 0$ and $p_k^j(\tau') > 0$ for all $\tau' \in \{\tau_1 + 1, \ldots, \tau_2 - 1\}$. The size of the epoch is given by $S = \sum_{i=\tau_1}^{\tau_2} p_k^j(i)$ and its corresponding duration given by $\tau_2 - \tau_1 - 1$. For a given ensemble of coarse-grained time series, all epochs were combined.

Scaling curve fit

For more precise evaluation of χ_{sh} and χ_{lg}, we introduced the following fitting function:

$$S(d) = \frac{C d^{\chi_{sh}}}{(1 + (d/\Phi)^{\gamma})^{(\chi_{sh}+\chi_{lg})/\gamma}}$$

This function is a double power law with initial slope χ_{sh}, transitioning to a second slope χ_{lg} at around the point $d = \Phi$. The parameter γ controls how abruptly that transition happens and has been fixed at 4 for all the curves presented. The other parameters were free to adjust to the data, and all fits were performed in log-space, i.e., the $\log(S(\log(d))$ was fit to the log of data (taking the log of both average sizes as well as durations).

Estimating inter-neuronal correlations during 'parabolic' and 'flat' avalanches

For each recording, we calculated the temporal coarse graining k, and scaling regime Φ for parabolic avalanches for which $\chi_{sh} \cong 2$. Durations of parabolic avalanches identified segments in the population raster over which pairwise correlations cc_{par} were obtained from all neuron pairs. Corresponding null-vectors were created by randomly selecting duration-matched time segments from 'flat' avalanches ($\chi_{lg} \cong 1$; n = 100 controls per experiment), from which cc_{flat} was calculated. Corresponding distributions of all *cc* for 'parabolic' and 'flat' avalanches were then compared.

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577

578 *Removal of inter-neuronal correlations by temporal shuffling*

579 Removal of inter-neuronal correlations was obtained by circular temporal shifts T of the entire
580 time series of individual neurons. T was chosen randomly between 0 and up to the full length of
581 the recording. These circular shifts maintain the precise temporal organization of each neuronal
582 time series. This approach isolates contributions from inter-neuronal correlations while
583 maintaining the first order statistics and sequences of inter spike intervals. For each percentage
584 of correlations removed, we averaged over n = 20 repeats for each k.

585

586 *Adding random spikes and random removal of cells*

587 *Random spike addition analysis*

588 To analyze the influence of uncorrelated activity on scaling trends, random spikes were added to
589 recording rasters after uncorrelated cell removal. This was done by adding spikes to the raster
590 randomly until the desired noise level was reached. We note that 100% noise addition
591 corresponds to doubling the firing rate in the raster.

592

593 *Cell removal and correlation removal analysis*

594 To analyze the robustness of scaling trends to cell and correlation removal, information was
595 removed from rasters progressively until the scaling trends were destroyed. For correlation
596 removal, a single ROI's activity in the raster was cyclically shifted by some random time T,
597 chosen uniformly from an interval between 0 and the length of the recording. This was repeated
598 for each ROI in the raster. For cell removal, an ROI was identified randomly and removed from

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the raster until the desired number of cells were removed. Both analyses were repeated 20 times and the scaling trends were averaged over.

Trial shuffling

Trial-shuffling for the GCaMP7s and GCaMP6f drifting grating evoked data was obtained by randomly permuting the responses from each of the 8 presented directions separately. This was done for each neuron independently. Therefore, in each trial of the trial-shuffled dataset activity from each cell corresponds to a response to the same stimulus presented in the original data, but taken from different presentations of that stimulus.

Bootstrapping

Robustness of avalanche scaling trends and temporal profiles was assessed using bootstrapping. For a given recording, avalanches were calculated with the normal preprocessing steps. The pool of avalanches was then resampled 10 times with replacement, such that each bootstrapped pool contained the same number of avalanches as the original pool. Error bars for each plot were reported as the standard deviation of the scaling trend or temporal profile across all bootstrapped pools.

State dependency of χ_{sh}

To study the effect of behavior state on network statistics and avalanche dynamics, spike rasters for rest and locomotion periods were extracted using wheel traces from individual recordings as follows ($N_{animals}$=4, $N_{recordings}$=7). First, the instantaneous wheel speed was binarized using a threshold of 0.25 cm/s. Then, using this vector as a mask, we obtained separate rest and

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locomotion spike rasters for each recording. Spike rate and pairwise correlation CDFs were computed on pooled data (Suppl. Fig. S7a, b) $<\lambda_{rest}>=0.44\pm0.26$Hz, $<\lambda_{loco}>=0.73\pm0.46$Hz, $p=0.00$; $<cc_{rest}>=0.033\pm0.033$, $<cc_{loco}>=0.030\pm0.028$, $p=0.00$). Avalanches for rest and locomotion were extracted as mentioned previously and pooled across animals and recordings. Threshold value increased as expected with increasing coarse graining factor ($<\Theta_{rest+loco}>=0.65\pm0.48$, $<\Theta_{rest}>=0.59\pm0.45$, $<\Theta_{loco}>=0.79\pm0.53$) and the number of avalanches varied as a function of threshold and behavior condition ($N_{rest+loco}=12804$, $N_{rest}=9766$, $N_{loco}=4165$ at native framerate). Finally, mean avalanche size vs. duration was calculated for each condition and scaling exponents, number of avalanches, and thresholds were extracted for different temporal coarse graining values of k (Suppl. Fig. S7c – f).

Temporal Profiles

Temporal profiles were calculated on individual recordings after thresholding and temporal coarse graining had been applied. Avalanches were obtained as described for our experimental data and grouped based on duration. Population activity during all avalanches of a certain duration was averaged over all avalanches to get the temporal profile for that duration. To calculate the profile collapse exponent χ^{coll}, temporal profiles were first x-rescaled to fit between 0 and 1, then linearly interpolated to N = 500 points. Every interpolated temporal profile in the desired duration range was then y-rescaled by $d^{\chi^{coll}-1}$. The exponent χ^{coll} was chosen to minimize the RMS error between all collapsed shapes in the desired duration range. To calculate temporal profiles at a certain coarse-graining k using the original experimental frame rate, avalanche epochs were calculated at the coarse-grained resolution as described above. Then, these epochs were mapped back onto the raster at the original frame rate and temporal profiles

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were calculated on these remapped epochs. The standard deviations from the mean rescaled shapes were visualized as shaded area if not stated otherwise.

Complexity analysis

Complexity analysis was performed as described in[39] exploring patterns of depths D in the range of 4 – 7. Complexity C was calculated on the population activity as a function of k and thresholding, identically to how epochs were computed to obtain scaling. Subthreshold activity was not evaluated. For each segment of length D, its pattern p is defined as the rank order of the time series (e.g., for a monotonically increasing trace over $D = 4$ time points, the pattern would be 0123; see Suppl. Fig. S15 for more examples). From all possible length D segments, a probability distribution $P \equiv \{p_j; j = 1, 2, \dots, N\}$ is obtained (N is the number of possible states, e.g., for $D = 3$, the possible states are 012, 021, 102, 120, 201 and 210). Next, the Shannon's logarithmic information is computed as $S[P] = -\sum_{j=1}^{N} p_j \ln(p_j)$, and the normalized Shannon entropy is defined as $H[P] = S[P]/S[P_e]$, where P_e is the uniform distribution (entropy is maximized for the uniform distribution, therefore $0 \leq H \leq 1$). Finally, the complexity measure is defined as $C[P] = Q_J[P, P_e] \times H[P]$, with $Q_J[P, P_e] = Q_0 \left(S\left[\frac{P+P_e}{2}\right] - \frac{S[P]}{2} - S[P_e]/2 \right)$. Q_0 is a normalization constant ($0 \leq Q_J \leq 1$) equal to the inverse of the maximum possible value of $J[P,P_e]$.

Neural simulations

Model topology

We adapted the model by Girardi-Schappo and colleagues[73], an excitatory/inhibitory (E/I) balanced system of integrate-and-fire (IF) neurons, which exhibits 4 domains of firing (low rate

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668 irregular, high rate, quiescent and intermittent) in addition to a 2nd-order, continuous phase
669 transition. At this phase transition, the model displays the dynamics of a critical branching
670 process with a slope of $\alpha = 3/2$ in avalanche size distribution and $\beta = 2$ for avalanche duration
671 distribution, which has been identified experimentally for neuronal avalanches[7]. Our standard
672 network consisted of $N = 10^6$ non-leaky, stochastic IF neurons. The E/I neuron ratio was set 4:1
673 to approximate prevalence of excitatory over inhibitory neurons in the cortex and neurons had an
674 all-to-all connectivity. The elements of the connectivity matrix, W, were initialized as $W^{EE} = W^{IE}$
675 $= J$; $W^{II} = W^{EI} = -gJ$, that is connections with an excitatory pre-synaptic neuron had synaptic
676 strength J and synapses with an inhibitory pre-synaptic neuron had synaptic strength -gJ. Here, g
677 is the synaptic balance parameter which was used to tune the model to an E/I-balanced state with
678 critical dynamics, and its critical value for the parameters of our network is, g_c=3.5 (Ref. [73]).
679 Values of g higher than g_c shift the network to an inhibition-dominated regime with subcritical
680 dynamics, and values of g lower than g_c shifts the network to an excitation-dominated regime
681 with supercritical dynamics.
682
683 *Model dynamics*. The state of a neuron in the model was described by 2 variables. A Boolean
684 variable, X, denotes a neuron's firing (or quiescence) at time t, i.e., $X(t) = 1$ (or $X(t) = 0$). The
685 membrane potential, V, controls the probability of firing and evolves as

686 $$V_i(t+1) = \mu V_i(t) + \sum_{j=1}^{N} W_{ij} X_j(t)\big(1 - X_j(t)\big),$$

687 where μ is a leakage parameter, which in our simulations was set to 0. The term $(1 - X_j(t))$
688 introduces an absolute refractory period of Δt as it resets the voltage after a spike, i.e., when $X_j(t)$
689 $= 1$. The probability of extra neuronal firing increased linearly with V according to

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690 $$P(X = 1 \mid V) = \begin{cases} V, \Gamma V < 1 \\ 1, \Gamma V \geq 1 \end{cases}$$

691 Where Γ is the neuronal gain, which in our simulation was set to 1 supporting probabilistic firing
692 near a 2nd-order phase transition[73]. This network is highly non-conservative with energy
693 dissipating through inhibition and spike collision.
694
695 *Model simulations*
696 The mean-field activity of the neurons show the hallmarks of neuronal avalanches with $\alpha = 3/2$
697 and $\beta = 2$ and $\chi = 2$ (Ref.[73]), which we confirmed in our finite-size, fully-sampled network (see
698 also Suppl. Fig. S10). Activity in the network is triggered by independent Poisson processes to
699 each neuron at low rate set to activate ~20 neurons on average (0.002% of the network) every
700 time step. This external driving triggers intermittent, reverberating activity in the network. For
701 our analysis, we thresholded the population activity above the expected average number of
702 spikes from the external drive to focus on the actual cascading activity in the network. Unless
703 stated otherwise, the network was simulated for 10^8 time-steps and the resulting time-series was
704 analyzed. Epochs were extracted as described for the data with identical size and duration
705 definition. We consider a random fraction *f* of the total neurons in the network. This sampling
706 fraction, *f*, was systematically varied from 0.01% (100 neurons) all the way to a fully sampled
707 network (10^6 neurons) with corresponding changes in Θ to study the effects of subsampling on

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the network activity. All curves of χ_{sh} and χ_{lg} for the simulated data use the scaling curve fit mentioned previously.

Statistics

All values are given as mean ± standard deviation (SD) if not stated otherwise. Violin plots were used to approximate the distribution of data (shaded area) bordered by the 1st and 3rd quartile. Box plots are composed of the median, box (25 – 75% quartiles) and whiskers (1.5x interquartile range).

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Authors Contributions

D.P. conceived and planned the study; P.K. and T.L.R. performed experiments; E.C., T.L.R., and D.P. took the lead in data analysis to which S.R.M. and D.R.C. contributed. K.S. performed the

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730 simulations. P.K. took the lead on deep-interpolation analysis. All authors contributed to
731 experimental aspects, discussion and writing of the manuscript.

732

733 Supplementary Information

734 See Supplementary Information

735 Competing interests

736 The authors declare no competing interests.

737

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